CHAPTER 6

DC-DC Converters

Dc-dc converters are power electronic circuits that convert a dc voltage to a different dc voltage level, often providing a regulated output. The circuits described in this chapter are classified as switched-mode dc-dc converters, also called switching power supplies or switchers. This chapter describes some basic dc-dc converter circuits. Chapter 7 describes some common variations of these circuits that are used in many dc power supply designs.

6.1 LINEAR VOLTAGE REGULATORS

Before we discuss switched-mode converters, it is useful to review the motivation for an alternative to linear dc-dc converters that was introduced in Chapt. 1. One method of converting a dc voltage to a lower dc voltage is a simple circuit as shown in Fig. 6-1. The output voltage is

$$V_o = I_L R_L$$

where the load current is controlled by the transistor. By adjusting the transistor base current, the output voltage may be controlled over a range of 0 to roughly V_s. The base current can be adjusted to compensate for variations in the supply voltage or the load, thus regulating the output. This type of circuit is called a linear dc-dc converter or a linear regulator because the transistor operates in the linear region, rather than in the saturation or cutoff regions. The transistor in effect operates as a variable resistance.

While this may be a simple way of converting a dc supply voltage to a lower dc voltage and regulating the output, the low efficiency of this circuit is a serious drawback for power applications. The power absorbed by the load is $V_o I_L$, and

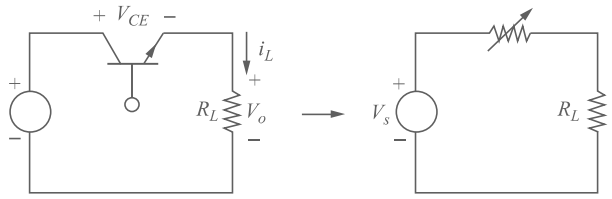


Figure 6-1 A basic linear regulator.

the power absorbed by the transistor is $V_{CE}I_L$, assuming a small base current. The power loss in the transistor makes this circuit inefficient. For example, if the output voltage is one-quarter of the input voltage, the load resistor absorbs one-quarter of the source power, which is an efficiency of 25 percent. The transistor absorbs the other 75 percent of the power supplied by the source. Lower output voltages result in even lower efficiencies. Therefore, the linear voltage regulator is suitable only for low-power applications.

6.2 A BASIC SWITCHING CONVERTER

An efficient alternative to the linear regulator is the switching converter. In a switching converter circuit, the transistor operates as an electronic switch by being completely on or completely off (saturation or cutoff for a BJT or the triode and cutoff regions of a MOSFET). This circuit is also known as a dc chopper.

Assuming the switch is ideal in Fig. 6-2, the output is the same as the input when the switch is closed, and the output is zero when the switch is open. Periodic

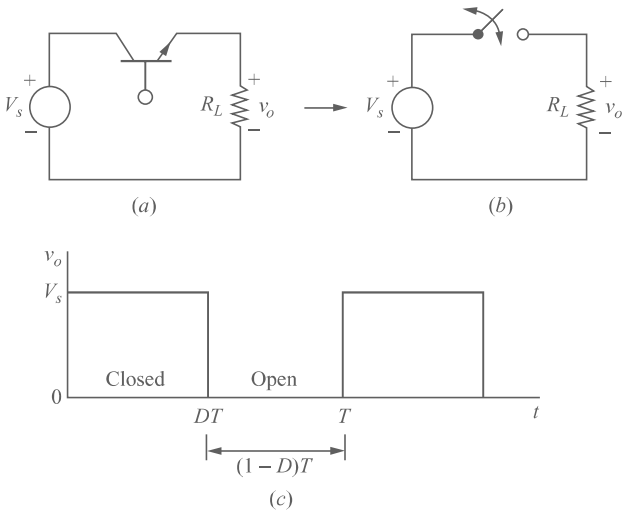


Figure 6-2 (*a*) A basic dc-dc switching converter; (*b*) Switching equivalent; (*c*) Output voltage.

opening and closing of the switch results in the pulse output shown in Fig. 6-2*c*. The average or dc component of the output voltage is

$$V_o = \frac{1}{T}\int_0^T v_o(t)\,dt = \frac{1}{T}\int_0^{DT} V_s\,dt = V_s D \tag{6-1}$$

The dc component of the output voltage is controlled by adjusting the duty ratio D, which is the fraction of the switching period that the switch is closed

$$D \equiv \frac{t_{\text{on}}}{t_{\text{on}} + t_{\text{off}}} = \frac{t_{\text{on}}}{T} = t_{\text{on}} f \tag{6-2}$$

where f is the switching frequency. The dc component of the output voltage will be less than or equal to the input voltage for this circuit.

The power absorbed by the ideal switch is zero. When the switch is open, there is no current in it; when the switch is closed, there is no voltage across it. Therefore, all power is absorbed by the load, and the energy efficiency is 100 percent. Losses will occur in a real switch because the voltage across it will not be zero when it is on, and the switch must pass through the linear region when making a transition from one state to the other.

6.3 THE BUCK (STEP-DOWN) CONVERTER

Controlling the dc component of a pulsed output voltage of the type in Fig. 6-2*c* may be sufficient for some applications, such as controlling the speed of a dc motor, but often the objective is to produce an output that is purely dc. One way of obtaining a dc output from the circuit of Fig. 6-2*a* is to insert a low-pass filter after the switch. Figure 6-3*a* shows an LC low-pass filter added to the basic converter. The diode provides a path for the inductor current when the switch is opened and is reverse-biased when the switch is closed. This circuit is called a *buck converter* or a *step-down converter* because the output voltage is less than the input.

Voltage and Current Relationships

If the low-pass filter is ideal, the output voltage is the average of the input voltage to the filter. The input to the filter, v_x in Fig. 6-3*a*, is V_s when the switch is closed and is zero when the switch is open, provided that the inductor current remains positive, keeping the diode on. If the switch is closed periodically at a duty ratio D, the average voltage at the filter input is $V_s D$, as in Eq. (6-1).

This analysis assumes that the diode remains forward-biased for the entire time when the switch is open, implying that the inductor current remains positive. An inductor current that remains positive throughout the switching period is known as *continuous current*. Conversely, discontinuous current is characterized by the inductor current's returning to zero during each period.

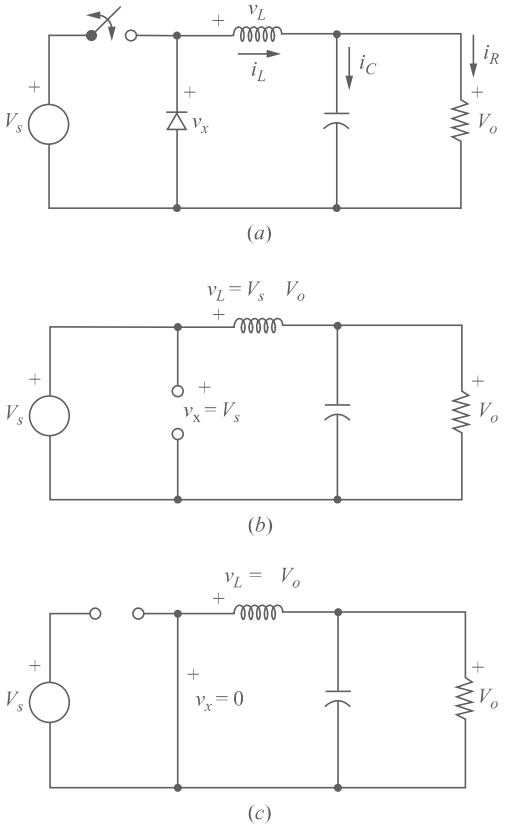


Figure 6-3 (*a*) Buck dc-dc converter; (*b*) Equivalent circuit for the switch closed; (*c*) Equivalent circuit for the switch open.

Another way of analyzing the operation of the buck converter of Fig. 6-3*a* is to examine the inductor voltage and current. This analysis method will prove useful for designing the filter and for analyzing circuits that are presented later in this chapter.

Buck converters and dc-dc converters in general, have the following properties when operating in the steady state:

1. The inductor current is periodic.

$$i_L(t + T) = i_L(t) \tag{6-3}$$

2. The average inductor voltage is zero (see Sec. 2.3).

$$V_L = \frac{1}{T}\int_{t}^{t+T} v_L(\lambda)d\lambda = 0 \tag{6-4}$$

3. The average capacitor current is zero (see Sec. 2.3).

$$I_C = \frac{1}{T}\int_{t}^{t+T} i_C(\lambda)\,d\lambda = 0 \tag{6-5}$$

4. The power supplied by the source is the same as the power delivered to the load. For nonideal components, the source also supplies the losses.

$$\begin{aligned} P_s &= P_o && \text{ideal} \\ P_s &= P_o + \text{losses} && \text{nonideal} \end{aligned} \tag{6-6}$$

Analysis of the buck converter of Fig. 6-3*a* begins by making these assumptions:

1. The circuit is operating in the steady state.
2. The inductor current is continuous (always positive).
3. The capacitor is very large, and the output voltage is held constant at voltage V_o. This restriction will be relaxed later to show the effects of finite capacitance.
4. The switching period is T; the switch is closed for time DT and open for time $(1-D)T$.
5. The components are ideal.

The key to the analysis for determining the output V_o is to examine the inductor current and inductor voltage first for the switch closed and then for the switch open. The net change in inductor current over one period must be zero for steady-state operation. The average inductor voltage is zero.

Analysis for the Switch Closed When the switch is closed in the buck converter circuit of Fig. 6-3*a*, the diode is reverse-biased and Fig. 6-3*b* is an equivalent circuit. The voltage across the inductor is

$$v_L = V_s - V_o = L\frac{di_L}{dt}$$

Rearranging,

$$\frac{di_L}{dt} = \frac{V_s - V_o}{L} \qquad \text{switch closed}$$

Since the derivative of the current is a positive constant, the current increases linearly as shown in Fig. 6-4*b*. The change in current while the switch is closed is computed by modifying the preceding equation.

$$\begin{aligned} \frac{di_L}{dt} &= \frac{\Delta i_L}{\Delta t} = \frac{\Delta i_L}{DT} = \frac{V_s - V_o}{L} \\ (\Delta i_L)_{\text{closed}} &= \left(\frac{V_s - V_o}{L}\right)DT \end{aligned} \tag{6-7}$$

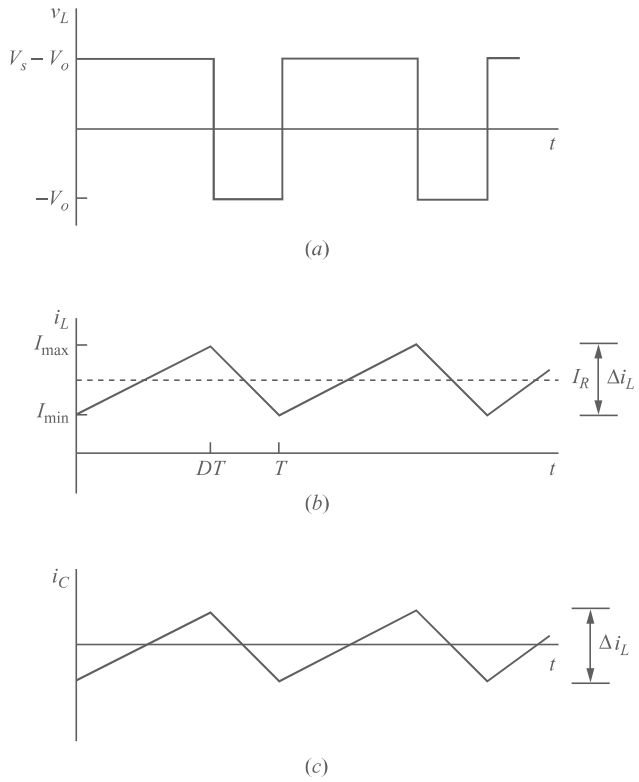


Figure 6-4 Buck converter waveforms: (*a*) Inductor voltage; (*b*) Inductor current; (*c*) Capacitor current.

Analysis for the Switch Open When the switch is open, the diode becomes forward-biased to carry the inductor current and the equivalent circuit of Fig. 6-3*c* applies. The voltage across the inductor when the switch is open is

$$v_L = -V_o = L\frac{di_L}{dt}$$

Rearranging,

$$\frac{di_L}{dt} = \frac{-V_o}{L} \qquad \text{switch open}$$

The derivative of current in the inductor is a negative constant, and the current decreases linearly as shown in Fig. 6-4*b*. The change in inductor current when the switch is open is

$$\frac{\Delta i_L}{\Delta t} = \frac{\Delta i_L}{(1-D)T} = -\frac{V_o}{L}$$
$$(\Delta i_L)_{\text{open}} = -\left(\frac{V_o}{L}\right)(1-D)\,T \tag{6-8}$$

Steady-state operation requires that the inductor current at the end of the switching cycle be the same as that at the beginning, meaning that the net change in inductor current over one period is zero. This requires

$$(\Delta i_L)_{\text{closed}} + (\Delta i_L)_{\text{open}} = 0$$

Using Eqs. (6-7) and (6-8),

$$\left(\frac{V_s - V_o}{L}\right)(DT) - \left(\frac{V_o}{L}\right)(1 - D)T = 0$$

Solving for V_o,

$$\boxed{V_o = V_s D} \tag{6-9}$$

which is the same result as Eq. (6-1). *The buck converter produces an output voltage that is less than or equal to the input.*

An alternative derivation of the output voltage is based on the inductor voltage, as shown in Fig. 6-4*a*. Since the average inductor voltage is zero for periodic operation,

$$V_L = (V_s - V_o)DT + (-V_o)(1 - D)T = 0$$

Solving the preceding equation for V_o yields the same result as Eq. (6-9), $V_o = V_s D$.

Note that the output voltage depends on only the input and the duty ratio D. If the input voltage fluctuates, the output voltage can be regulated by adjusting the duty ratio appropriately. A feedback loop is required to sample the output voltage, compare it to a reference, and set the duty ratio of the switch accordingly. Regulation techniques are discussed in Chap. 7.

The average inductor current must be the same as the average current in the load resistor, since the average capacitor current must be zero for steady-state operation:

$$I_L = I_R = \frac{V_o}{R} \tag{6-10}$$

Since the change in inductor current is known from Eqs. (6-7) and (6-8), the maximum and minimum values of the inductor current are computed as

$$\begin{aligned} I_{\max} &= I_L + \frac{\Delta i_L}{2} \\ &= \frac{V_o}{R} + \frac{1}{2}\left[\frac{V_o}{L}(1 - D)T\right] = V_o\left(\frac{1}{R} + \frac{1 - D}{2Lf}\right) \end{aligned} \tag{6-11}$$

$$\begin{aligned} I_{\min} &= I_L - \frac{\Delta i_L}{2} \\ &= \frac{V_o}{R} - \frac{1}{2}\left[\frac{V_o}{L}(1 - D)T\right] = V_o\left(\frac{1}{R} - \frac{1 - D}{2Lf}\right) \end{aligned} \tag{6-12}$$

where $f = 1/T$ is the switching frequency.

For the preceding analysis to be valid, continuous current in the inductor must be verified. An easy check for continuous current is to calculate the minimum inductor current from Eq. (6-12). Since the minimum value of inductor current must be positive for continuous current, a negative minimum calculated from Eq. (6-12) is not allowed due to the diode and indicates discontinuous current. The circuit will operate for discontinuous inductor current, but the preceding analysis is not valid. Discontinuous-current operation is discussed later in this chapter.

Equation (6-12) can be used to determine the combination of L and f that will result in continuous current. Since $I_{\min} = 0$ is the boundary between continuous and discontinuous current,

$$I_{\min} = 0 = V_o\left(\frac{1}{R} - \frac{1-D}{2Lf}\right)$$
$$(Lf)_{\min} = \frac{(1-D)R}{2} \tag{6-13}$$

If the desired switching frequency is established,

$$\boxed{L_{\min} = \frac{(1-D)R}{2f} \qquad \text{for continuous current}} \tag{6-14}$$

where $L_{\min}$ is the minimum inductance required for continuous current. In practice, a value of inductance greater than $L_{\min}$ is desirable to ensure continuous current.

In the design of a buck converter, the peak-to-peak variation in the inductor current is often used as a design criterion. Equation (6-7) can be combined with Eq. (6-9) to determine the value of inductance for a specified peak-to-peak inductor current for continuous-current operation:

$$\Delta i_L = \left(\frac{V_s - V_o}{L}\right)DT = \left(\frac{V_s - V_o}{Lf}\right)D = \frac{V_o(1-D)}{Lf} \tag{6-15}$$

or

$$\boxed{L = \left(\frac{V_s - V_o}{\Delta i_L f}\right)D = \frac{V_o(1-D)}{\Delta i_L f}} \tag{6-16}$$

Since the converter components are assumed to be ideal, the power supplied by the source must be the same as the power absorbed by the load resistor.

$$P_s = P_o$$
$$V_s I_s = V_o I_o \tag{6-17}$$

or

$$\frac{V_o}{V_s} = \frac{I_s}{I_o}$$

Note that the preceding relationship is similar to the voltage-current relationship for a transformer in ac applications. Therefore, the buck converter circuit is equivalent to a dc transformer.

Output Voltage Ripple

In the preceding analysis, the capacitor was assumed to be very large to keep the output voltage constant. In practice, the output voltage cannot be kept perfectly constant with a finite capacitance. The variation in output voltage, or ripple, is computed from the voltage-current relationship of the capacitor. The current in the capacitor is

$$i_C = i_L - i_R$$

shown in Fig. 6-5*a*.

While the capacitor current is positive, the capacitor is charging. From the definition of capacitance,

$$Q = CV_o$$

$$\Delta Q = C\Delta V_o$$

$$\Delta V_o = \frac{\Delta Q}{C}$$

The change in charge ΔQ is the area of the triangle above the time axis

$$\Delta Q = \frac{1}{2}\left(\frac{T}{2}\right)\left(\frac{\Delta i_L}{2}\right) = \frac{T\Delta i_L}{8}$$

resulting in

$$\Delta V_o = \frac{T\Delta i_L}{8C}$$

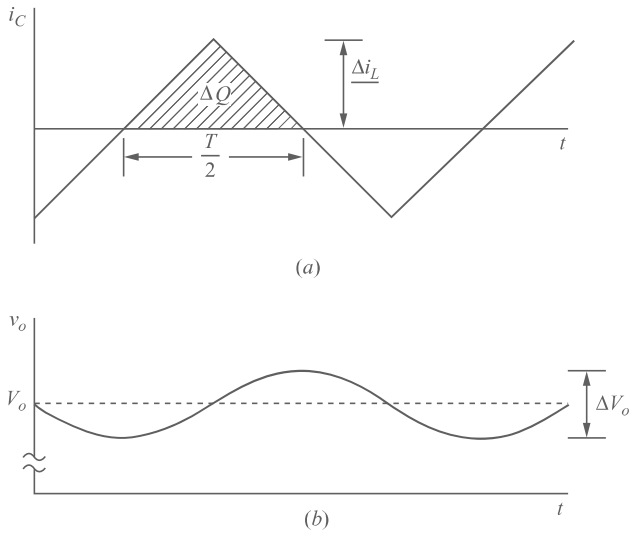


Figure 6-5 Buck converter waveforms. (*a*) Capacitor current; (*b*) Capacitor ripple voltage.

Using Eq. (6-8) for Δi_L,

$$\Delta V_o = \frac{T V_o}{8CL}(1 - D)T = \frac{V_o(1 - D)}{8LCf^2} \tag{6-18}$$

In this equation, ΔV_o is the peak-to-peak ripple voltage at the output, as shown in Fig. 6-5*b*. It is also useful to express the ripple as a fraction of the output voltage,

$$\boxed{\frac{\Delta V_o}{V_o} = \frac{1 - D}{8LCf^2}} \tag{6-19}$$

In design, it is useful to rearrange the preceding equation to express required capacitance in terms of specified voltage ripple:

$$\boxed{C = \frac{1 - D}{8L(\Delta V_o/V_o)f^2}} \tag{6-20}$$

If the ripple is not large, the assumption of a constant output voltage is reasonable and the preceding analysis is essentially valid.

EXAMPLE 6-1

Buck Converter

The buck dc-dc converter of Fig. 6-3*a* has the following parameters:

$V_s = 50$ V
$D = 0.4$
$L = 400$ μH
$C = 100$ μF
$f = 20$ kHz
$R = 20$ Ω

Assuming ideal components, calculate (*a*) the output voltage V_o, (*b*) the maximum and minimum inductor current, and (*c*) the output voltage ripple.

■ Solution

(a) The inductor current is assumed to be continuous, and the output voltage is computed from Eq. (6-9),

$$V_o = V_s D = (50)(0.4) = 20 \text{ V}$$

(b) Maximum and minimum inductor currents are computed from Eqs. (6-11) and (6-12).

$$I_{\max} = V_o\left(\frac{1}{R} + \frac{1 - D}{2Lf}\right)$$

$$= 20\left[\frac{1}{20} + \frac{1 - 0.4}{2(400)(10)^{-6}(20)(10)^3}\right]$$

$$= 1 + \frac{1.5}{2} = 1.75 \text{ A}$$

$$I_{\min} = V_o\left(\frac{1}{R} - \frac{1-D}{2Lf}\right)$$
$$= 1 - \frac{1.5}{2} = 0.25 \text{ A}$$

The average inductor current is 1 A, and $\Delta i_L = 1.5$ A. Note that the minimum inductor current is positive, verifying that the assumption of continuous current was valid.

(c) The output voltage ripple is computed from Eq. (6-19).

$$\frac{\Delta V_o}{V_o} = \frac{1-D}{8LCf^2} = \frac{1-0.4}{8(400)(10)^{-6}(100)(10)^{-6}(20{,}000)^2}$$
$$= 0.00469 = 0.469\%$$

Since the output ripple is sufficiently small, the assumption of a constant output voltage was reasonable.

Capacitor Resistance—The Effect on Ripple Voltage

The output voltage ripple in Eqs. (6-18) and (6-19) is based on an ideal capacitor. A real capacitor can be modeled as a capacitance with an equivalent series resistance (ESR) and an equivalent series inductance (ESL). The ESR may have a significant effect on the output voltage ripple, often producing a ripple voltage greater than that of the ideal capacitance. The inductance in the capacitor is usually not a significant factor at typical switching frequencies. Figure 6.6 shows a capacitor model that is appropriate for most applications.

The ripple due to the ESR can be approximated by first determining the current in the capacitor, assuming the capacitor to be ideal. For the buck converter in the continuous-current mode, capacitor current is the triangular current waveform of Fig. 6-4*c*. The voltage variation across the capacitor resistance is

$$\Delta V_{o,\text{ESR}} = \Delta i_C r_C = \Delta i_L r_C \tag{6-21}$$

To estimate a worst-case condition, one could assume that the peak-to-peak ripple voltage due to the ESR algebraically adds to the ripple due to the capacitance. However, the peaks of the capacitor and the ESR ripple voltages will not coincide, so

$$\Delta V_o < \Delta V_{o,C} + \Delta V_{o,\text{ESR}} \tag{6-22}$$

where $\Delta V_{o,C}$ is ΔV_o in Eq. (6-18). The ripple voltage due to the ESR can be much larger than the ripple due to the pure capacitance. In that case, the output capacitor is chosen on the basis of the equivalent series resistance rather than capacitance only.

$$\Delta V_o \approx \Delta V_{o,\text{ESR}} = \Delta i_C r_C \tag{6-23}$$



Figure 6-6 A model for the capacitor including the equivalent series resistance (ESR).

Capacitor ESR is inversely proportional to the capacitance value—a larger capacitance results in a lower ESR. Manufacturers provide what are known as *low-ESR capacitors* for power supply applications.

In Example 6-1, the 100-μF capacitor may have an ESR of $r_C = 0.1\ \Omega$. The ripple voltage due to the ESR is calculated as

$$\Delta V_{o,\text{ESR}} = \Delta i_C r_C = \Delta i_L r_C = (1.5\,\text{A})(0.1\ \Omega) = 0.15\,\text{V}$$

Expressed as a percent, $\Delta V_o/V_o$ is 0.15/20 = 0.75 percent. The total ripple can then be approximated as 0.75 percent.

Synchronous Rectification for the Buck Converter

Many buck converters use a second MOSFET in place of the diode. When S_2 is on and S_1 is off, current flows upward out of the drain of S_2. The advantage of this configuration is that the second MOSFET will have a much lower voltage drop across it compared to a diode, resulting in higher circuit efficiency. This is especially important in low-voltage, high-current applications. A Shottky diode would have a voltage of 0.3 to 0.4 V across it while conducting, whereas a MOSFET will have an extremely low voltage drop due to an $R_{DS_{\text{on}}}$ as low as single-digit milliohms. This circuit has a control scheme known as *synchronous switching*, or *synchronous rectification*. The second MOSFET is known as a *synchronous rectifier*. The two MOSFETs must not be on at the same time to prevent a short circuit across the source, so a "dead time" is built into the switching control—one MOSFET is turned off before the other is turned on. A diode is placed in parallel with the second MOSFET to provide a conducting path for inductor current during the dead time when both MOSFETs are off. This diode may be the MOSFET body diode, or it may be an extra diode, most likely a Shottky diode, for improved switching. The synchronous buck converter should be operated in the continuous-current mode because the MOSFET would allow the inductor current to go negative.

Other converter topologies presented in this chapter and in Chap. 7 can utilize MOSFETs in place of diodes.

6.4 DESIGN CONSIDERATIONS

Most buck converters are designed for continuous-current operation. The choice of switching frequency and inductance to give continuous current is given by Eq. (6-13), and the output voltage ripple is described by Eqs. (6-16) and (6-21). Note that as the switching frequency increases, the minimum size of the inductor to produce continuous current and the minimum size of the capacitor to limit output ripple both decrease. Therefore, high switching frequencies are desirable to reduce the size of both the inductor and the capacitor.

The tradeoff for high switching frequencies is increased power loss in the switches, which is discussed later in this chapter and in Chap. 10. Increased power loss in the switches means that heat is produced. This decreases the converter's efficiency and may require a large heat sink, offsetting the reduction in size of the inductor and capacitor. Typical switching frequencies are above 20 kHz to avoid audio noise, and they extend well into the 100s of kilohertz and into the megahertz range. Some designers consider about 500 kHz to be the best compromise

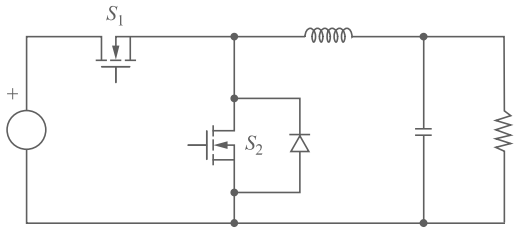


Figure 6-7 A synchronous buck converter. The MOSFET S_2 carries the inductor current when S_1 is off to provide a lower voltage drop than a diode.

between small component size and efficiency. Other designers prefer to use lower switching frequencies of about 50 kHz to keep switching losses small, while still others prefer frequencies larger than 1 MHz. As switching devices improve, switching frequencies will increase.

For low-voltage, high-current applications, the synchronous rectification scheme of Fig. 6-7 is preferred over using a diode for the second switch. The voltage across the conducting MOSFET will be much less than that across a diode, resulting in lower losses.

The inductor value should be larger than L_{min} in Eq. (6-14) to ensure continuous-current operation. Some designers select a value 25 percent larger than L_{min}. Other designers use different criteria, such as setting the inductor current variation, Δi_L in Eq. (6-15), to a desired value, such as 40 percent of the average inductor current. A smaller Δi_L results in lower peak and rms inductor currents and a lower rms capacitor current but requires a larger inductor.

The inductor wire must be rated at the rms current, and the core should not saturate for peak inductor current. The capacitor must be selected to limit the output ripple to the design specifications, to withstand peak output voltage, and to carry the required rms current.

The switch (usually a MOSFET with a low $R_{DS_{\text{on}}}$) and diode (or second MOSFET for synchronous rectification) must withstand maximum voltage stress when off and maximum current when on. The temperature ratings must not be exceeded, often requiring a heat sink.

Assuming ideal switches and an ideal inductor in the initial design is usually reasonable. However, the ESR of the capacitor should be included because it typically gives a more significant output voltage ripple than the ideal device and greatly influences the choice of capacitor size.

EXAMPLE 6-2

Buck Converter Design 1

Design a buck converter to produce an output voltage of 18 V across a 10-Ω load resistor. The output voltage ripple must not exceed 0.5 percent. The dc supply is 48 V. Design for continuous inductor current. Specify the duty ratio, the switching frequency, the values of the inductor and capacitor, the peak voltage rating of each device, and the rms current in the inductor and capacitor. Assume ideal components.

Solution

Using the buck converter circuit in Fig. 6-3*a*, the duty ratio for continuous-current operation is determined from Eq. (6-9):

$$D = \frac{V_o}{V_s} = \frac{18}{48} = 0.375$$

The switching frequency and inductor size must be selected for continuous-current operation. Let the switching frequency arbitrarily be 40 kHz, which is well above the audio range and is low enough to keep switching losses small. The minimum inductor size is determined from Eq. (6-14).

$$L_{\min} = \frac{(1 - D)(R)}{2f} = \frac{(1 - 0.375)(10)}{2(40{,}000)} = 78\,\mu\text{H}$$

Let the inductor be 25 percent larger than the minimum to ensure that inductor current is continuous.

$$L = 1.25L_{\min} = (1.25)(78\ \mu\text{H}) = 97.5\,\mu\text{H}$$

Average inductor current and the change in current are determined from Eqs. (6-10) and (6-17).

$$I_L = \frac{V_o}{R} = \frac{18}{10} = 1.8\ \text{A}$$

$$\Delta i_L = \left(\frac{V_s - V_o}{L}\right)DT = \frac{48 - 18}{97.5(10)^{-6}}(0.375)\left(\frac{1}{40{,}000}\right) = 2.88\ \text{A}$$

The maximum and minimum inductor currents are determined from Eqs. (6-11) and (6-12).

$$I_{\max} = I_L + \frac{\Delta i_L}{2} = 1.8 + 1.44 = 3.24\ \text{A}$$

$$I_{\min} = I_L - \frac{\Delta i_L}{2} = 1.8 - 1.44 = 0.36\ \text{A}$$

The inductor must be rated for rms current, which is computed as in Chap. 2 (see Example 2-8). For the offset triangular wave,

$$I_{L,\text{rms}} = \sqrt{I_L^2 + \left(\frac{\Delta i_L/2}{\sqrt{3}}\right)^2} = \sqrt{(1.8)^2 + \left(\frac{1.44}{\sqrt{3}}\right)^2} = 1.98\ \text{A}$$

The capacitor is selected using Eq. (6-20).

$$C = \frac{1 - D}{8L(\Delta V_o/V_o)f^2} = \frac{1 - 0.375}{8(97.5)(10)^{-6}(0.005)(40{,}000)^2} = 100\,\mu\text{F}$$

Peak capacitor current is $\Delta i_L/2 = 1.44$ A, and rms capacitor current for the triangular waveform is $1.44/\sqrt{3} = 0.83$ A. The maximum voltage across the switch and diode is V_s, or 48 V. The inductor voltage when the switch is closed is $V_s - V_o = 48 - 18 = 30$ V. The inductor voltage when the switch is open is $V_o = 18$ V. Therefore, the inductor must withstand 30 V. The capacitor must be rated for the 18-V output.

EXAMPLE 6-3

Buck Converter Design 2

Power supplies for telecommunications applications may require high currents at low voltages. Design a buck converter that has an input voltage of 3.3 V and an output voltage of 1.2 V. The output current varies between 4 and 6 A. The output voltage ripple must not exceed 2 percent. Specify the inductor value such that the peak-to-peak variation in inductor current does not exceed 40 percent of the average value. Determine the required rms current rating of the inductor and of the capacitor. Determine the maximum equivalent series resistance of the capacitor.

■ Solution

Because of the low voltage and high output current in this application, the synchronous rectification buck converter of Fig. 6-7 is used. The duty ratio is determined from Eq. (6-9).

$$D = \frac{V_o}{V_s} = \frac{1.2}{3.3} = 0.364$$

The switching frequency and inductor size must be selected for continuous-current operation. Let the switching frequency arbitrarily be 500 kHz to give a good tradeoff between small component size and low switching losses.

The average inductor current is the same as the output current. Analyzing the circuit for an output current of 4 A,

$$I_L = I_o = 4\,\text{A}$$

$$\Delta i_L = (40\%)(4) = 1.6\,\text{A}$$

Using Eq. (6-16),

$$L = \left(\frac{V_s - V_o}{\Delta i_L f}\right)\text{D} = \frac{3.3 - 1.2}{(1.6)(500{,}000)}(0.364) = 0.955\ \mu\text{H}$$

Analyzing the circuit for an output current of 6 A,

$$I_L = I_o = 6\,\text{A}$$

$$\Delta i_L = (40\%)(6) = 2.4\,\text{A}$$

resulting in

$$L = \left(\frac{V_s - V_o}{\Delta i_L f}\right)D = \frac{3.3 - 1.2}{(2.4)(500{,}000)}(0.364) = 0.636\ \mu\text{H}$$

Since 0.636 μH would be too small for the 4-A output, use $L = 0.955$ μH, which would be rounded to 1 μH.

Inductor rms current is determined from

$$I_{L,\text{rms}} = \sqrt{I_L^2 + \left(\frac{\Delta i_L/2}{\sqrt{3}}\right)^2}$$

(see Chap. 2). From Eq. (6-15), the variation in inductor current is 1.6 A for each output current. Using the 6-A output current, the inductor must be rated for an rms current of

$$I_{L,\text{rms}} = \sqrt{6^2 + \left(\frac{0.8}{\sqrt{3}}\right)^2} = 6.02 \text{ A}$$

Note that the average inductor current would be a good approximation to the rms current since the variation is relatively small.

Using $L = 1$ μH in Eq. (6-20), the minimum capacitance is determined as

$$C = \frac{1 - D}{8L(\Delta V_o / V_o)f^2} = \frac{1 - 0.364}{8(1)(10)^{-6}(0.02)(500{,}000)^2} = 0.16 \text{ μF}$$

The allowable output voltage ripple of 2 percent is (0.02)(1.2) = 24 mV. The maximum ESR is computed from Eq. (6-23).

$$\Delta V_o \approx r_C \Delta i_C = r_C \Delta i_L$$

or

$$r_C = \frac{\Delta V_o}{\Delta i_C} = \frac{24 \text{ mV}}{1.6 \text{ A}} = 15 \text{ mΩ}$$

At this point, the designer would search manufacturer's specifications for a capacitor having 15-mΩ ESR. The capacitor may have to be much larger than the calculated value of 0.16 μF to meet the ESR requirement. Peak capacitor current is $\Delta i_L/2 = 0.8$ A, and rms capacitor current for the triangular waveform is $0.8/\sqrt{3} = 0.46$ A.

6.5 THE BOOST CONVERTER

The boost converter is shown in Fig. 6-8. This is another switching converter that operates by periodically opening and closing an electronic switch. It is called a boost converter because the output voltage is larger than the input.

Voltage and Current Relationships

The analysis assumes the following:

1. Steady-state conditions exist.
2. The switching period is T, and the switch is closed for time DT and open for $(1-D)T$.
3. The inductor current is continuous (always positive).
4. The capacitor is very large, and the output voltage is held constant at voltage V_o.
5. The components are ideal.

The analysis proceeds by examining the inductor voltage and current for the switch closed and again for the switch open.

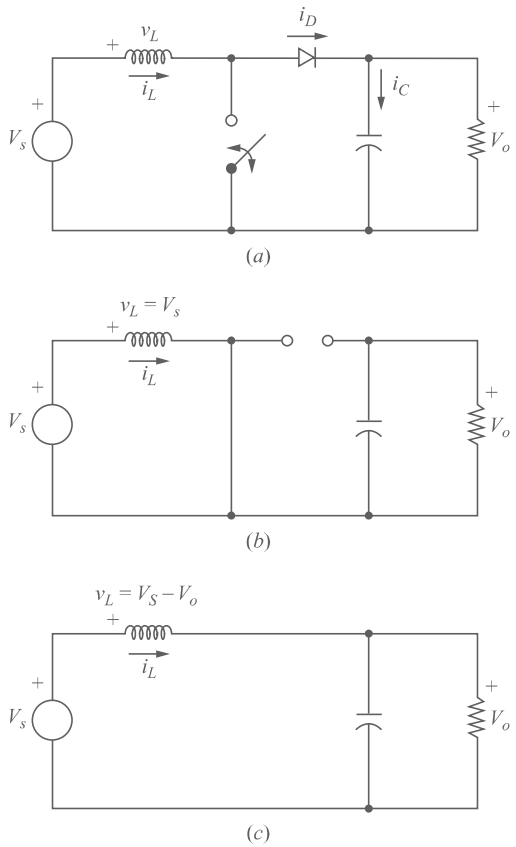


Figure 6-8 The boost converter. (*a*) Circuit; (*b*) Equivalent circuit for the switch closed; (*c*) Equivalent circuit for the switch open.

Analysis for the Switch Closed When the switch is closed, the diode is reverse-biased. Kirchhoff's voltage law around the path containing the source, inductor, and closed switch is

$$v_L = V_s = L\frac{di_L}{dt} \qquad \text{or} \qquad \frac{di_L}{dt} = \frac{V_s}{L} \tag{6-24}$$

The rate of change of current is a constant, so the current increases linearly while the switch is closed, as shown in Fig. 6-9*b*. The change in inductor current is computed from

$$\frac{\Delta i_L}{\Delta t} = \frac{\Delta i_L}{DT} = \frac{V_s}{L}$$

Solving for Δi_L for the switch closed,

$$(\Delta i_L)_{\text{closed}} = \frac{V_s DT}{L} \tag{6-25}$$

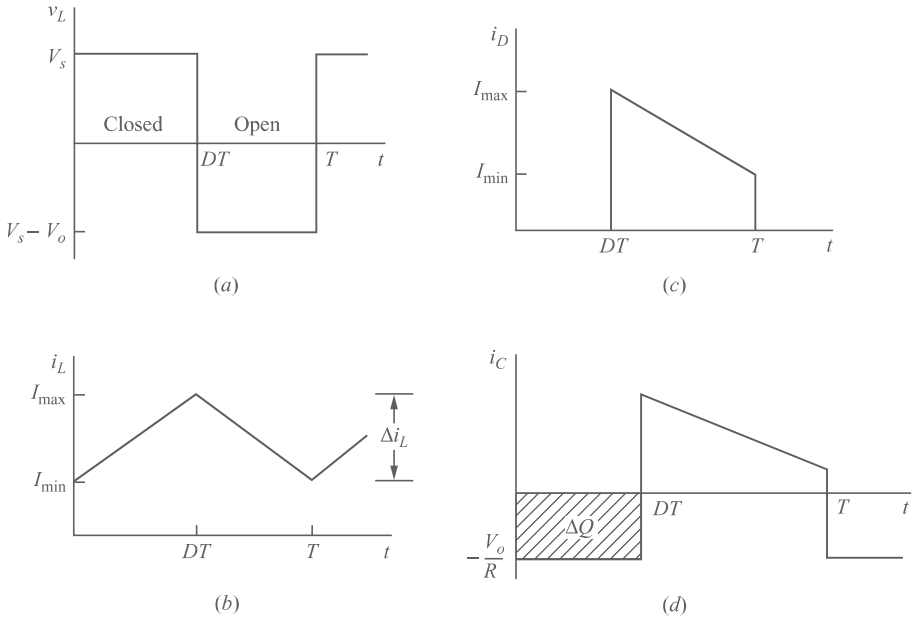


Figure 6-9 Boost converter waveforms. (*a*) Inductor voltage; (*b*) Inductor current; (*c*) Diode current; (*d*) Capacitor current.

Analysis for the Switch Open When the switch is opened, the inductor current cannot change instantaneously, so the diode becomes forward-biased to provide a path for inductor current. Assuming that the output voltage V_o is a constant, the voltage across the inductor is

$$v_L = V_s - V_o = L\frac{di_L}{dt}$$

$$\frac{di_L}{dt} = \frac{V_s - V_o}{L}$$

The rate of change of inductor current is a constant, so the current must change linearly while the switch is open. The change in inductor current while the switch is open is

$$\frac{\Delta i_L}{\Delta t} = \frac{\Delta i_L}{(1-D)T} = \frac{V_s - V_o}{L}$$

Solving for Δi_L,

$$(\Delta i_L)_{\text{open}} = \frac{(V_s - V_o)(1-D)T}{L} \tag{6-26}$$

For steady-state operation, the net change in inductor current must be zero. Using Eqs. (6-25) and (6-26),

$$(\Delta i_L)_{\text{closed}} + (\Delta i_L)_{\text{open}} = 0$$

$$\frac{V_s DT}{L} + \frac{(V_s - V_o)(1 - D)T}{L} = 0$$

Solving for V_o,

$$V_s(D + 1 - D) - V_o(1 - D) = 0$$

$$\boxed{V_o = \frac{V_s}{1 - D}} \tag{6-27}$$

Also, the average inductor voltage must be zero for periodic operation. Expressing the average inductor voltage over one switching period,

$$V_L = V_s D + (V_s - V_o)(1 - D) = 0$$

Solving for V_o yields the same result as in Eq. (6-27).

Equation (6-27) shows that if the switch is always open and D is zero, the output voltage is the same as the input. As the duty ratio is increased, the denominator of Eq. (6-27) becomes smaller, resulting in a larger output voltage. *The boost converter produces an output voltage that is greater than or equal to the input voltage.* However, the output voltage cannot be less than the input, as was the case with the buck converter.

As the duty ratio of the switch approaches 1, the output voltage goes to infinity according to Eq. (6-27). However, Eq. (6-27) is based on ideal components. Real components that have losses will prevent such an occurrence, as shown later in this section. Figure 6-9 shows the voltage and current waveforms for the boost converter.

The average current in the inductor is determined by recognizing that the average power supplied by the source must be the same as the average power absorbed by the load resistor. Output power is

$$P_o = \frac{V_o^2}{R} = V_o I_o$$

and input power is $V_s I_s = V_s I_L$. Equating input and output powers and using Eq. (6-27),

$$V_s I_L = \frac{V_o^2}{R} = \frac{[V_s/(1 - D)]^2}{R} = \frac{V_s^2}{(1 - D)^2} R$$

By solving for average inductor current and making various substitutions, I_L can be expressed as

$$I_L = \frac{V_s}{(1-D)^2 R} = \frac{V_o^2}{V_s R} = \frac{V_o I_o}{V_s} \tag{6-28}$$

Maximum and minimum inductor currents are determined by using the average value and the change in current from Eq. (6-25).

$$I_{\max} = I_L + \frac{\Delta i_L}{2} = \frac{V_s}{(1-D)^2 R} + \frac{V_s DT}{2L} \tag{6-29}$$

$$I_{\min} = I_L - \frac{\Delta i_L}{2} = \frac{V_s}{(1-D)^2 R} - \frac{V_s DT}{2L} \tag{6-30}$$

Equation (6-27) was developed with the assumption that the inductor current is continuous, meaning that it is always positive. A condition necessary for continuous inductor current is for $I_{\min}$ to be positive. Therefore, the boundary between continuous and discontinuous inductor current is determined from

$$I_{\min} = 0 = \frac{V_s}{(1-D)^2 R} - \frac{V_s DT}{2L}$$

or

$$\frac{V_s}{(1-D)^2 R} = \frac{V_s DT}{2L} = \frac{V_s D}{2Lf}$$

The minimum combination of inductance and switching frequency for continuous current in the boost converter is therefore

$$(Lf)_{\min} = \frac{D(1-D)^2 R}{2} \tag{6-31}$$

or

$$L_{\min} = \frac{D(1-D)^2 R}{2f} \tag{6-32}$$

A boost converter designed for continuous-current operation will have an inductor value greater than $L_{\min}$.

From a design perspective, it is useful to express L in terms of a desired Δi_L,

$$L = \frac{V_s DT}{\Delta i_L} = \frac{V_s D}{\Delta i_L f} \tag{6-33}$$

Output Voltage Ripple

The preceding equations were developed on the assumption that the output voltage was a constant, implying an infinite capacitance. In practice, a finite capacitance will result in some fluctuation in output voltage, or ripple.

The peak-to-peak output voltage ripple can be calculated from the capacitor current waveform, shown in Fig. 6-9*d*. The change in capacitor charge can be calculated from

$$|\Delta Q| = \left(\frac{V_o}{R}\right) DT = C\Delta V_o$$

An expression for ripple voltage is then

$$\Delta V_o = \frac{V_o DT}{RC} = \frac{V_o D}{RCf}$$

or

$$\boxed{\frac{\Delta V_o}{V_o} = \frac{D}{RCf}} \tag{6-34}$$

where f is the switching frequency. Alternatively, expressing capacitance in terms of output voltage ripple yields

$$C = \frac{D}{R(\Delta V_o / V_o) f} \tag{6-35}$$

As with the buck converter, equivalent series resistance of the capacitor can contribute significantly to the output voltage ripple. The peak-to-peak variation in capacitor current (Fig. 6-9) is the same as the maximum current in the inductor. The voltage ripple due to the ESR is

$$\Delta V_{o,\text{ESR}} = \Delta i_C r_C = I_{L,\max} r_C \tag{6-36}$$

EXAMPLE 6-4

Boost Converter Design 1

Design a boost converter that will have an output of 30 V from a 12-V source. Design for continuous inductor current and an output ripple voltage of less than one percent. The load is a resistance of 50 Ω. Assume ideal components for this design.

■ Solution

First, determine the duty ratio from Eq. (6-27),

$$D = 1 - \frac{V_s}{V_o} = 1 - \frac{12}{30} = 0.6$$

If the switching frequency is selected at 25 kHz to be above the audio range, then the minimum inductance for continuous current is determined from Eq. (6-32).

$$L_{\min} = \frac{D(1-D)^2(R)}{2f} = \frac{0.6(1-0.6)^2(50)}{2(25{,}000)} = 96\ \mu\text{H}$$

To provide a margin to ensure continuous current, let $L = 120\ \mu$H. Note that L and f are selected somewhat arbitrarily and that other combinations will also give continuous current.

Using Eqs. (6-28) and (6-25),

$$I_L = \frac{V_s}{(1-D)^2(R)} = \frac{12}{(1-0.6)^2(50)} = 1.5\ \text{A}$$

$$\frac{\Delta i_L}{2} = \frac{V_s DT}{2L} = \frac{(12)(0.6)}{(2)(120)(10)^{-6}(25{,}000)} = 1.2\ \text{A}$$

$$I_{\max} = 1.5 + 1.2 = 2.7\,\text{A}$$

$$I_{\min} = 1.5 - 1.2 = 0.3\,\text{A}$$

The minimum capacitance required to limit the output ripple voltage to 1 percent is determined from Eq. (6-35).

$$C \geq \frac{D}{R(\Delta V_o / V_o) f} = \frac{0.6}{(50)(0.01)(25{,}000)} = 48\ \mu\text{F}$$

EXAMPLE 6-5

Boost Converter Design 2

A boost converter is required to have an output voltage of 8 V and supply a load current of 1 A. The input voltage varies from 2.7 to 4.2 V. A control circuit adjusts the duty ratio to keep the output voltage constant. Select the switching frequency. Determine a value for the inductor such that the variation in inductor current is no more than 40 percent of the average inductor current for all operating conditions. Determine a value of an ideal capacitor such that the output voltage ripple is no more than 2 percent. Determine the maximum capacitor equivalent series resistance for a 2 percent ripple.

■ Solution

Somewhat arbitrarily, choose 200 kHz for the switching frequency. The circuit must be analyzed for both input voltage extremes to determine the worst-case condition. For $V_s = 2.7$ V, the duty ratio is determined from Eq. (6-27).

$$D = 1 - \frac{V_s}{V_o} = 1 - \frac{2.7}{8} = 0.663$$

Average inductor current is determined from Eq. (6-28).

$$I_L = \frac{V_o I_o}{V_s} = \frac{8(1)}{2.7} = 2.96\ \text{A}$$

The variation in inductor current to meet the 40 percent specification is then $\Delta i_L = 0.4(2.96) = 1.19$ A. The inductance is then determined from Eq. (6-33).

$$L = \frac{V_s D}{\Delta i_L f} = \frac{2.7(0.663)}{1.19(200{,}000)} = 7.5\ \mu\text{H}$$

Repeating the calculations for $V_s = 4.2$V,

$$D = 1 - \frac{V_s}{V_o} = 1 - \frac{4.2}{8} = 0.475$$

$$I_L = \frac{V_o I_o}{V_s} = \frac{8(1)}{4.2} = 1.90\ \text{A}$$

The variation in inductor current for this case is $\Delta i_L = 0.4(1.90) = 0.762$ A, and

$$L = \frac{V_s D}{\Delta i_L f} = \frac{4.2(0.475)}{0.762(200{,}000)} = 13.1\ \mu\text{H}$$

The inductor must be 13.1 μH to satisfy the specifications for the total range of input voltages.

Equation (6-35), using the maximum value of D, gives the minimum capacitance as

$$C = \frac{D}{R(\Delta V_o/V_o)f} = \frac{D}{(V_o/I_o)(\Delta V_o/V_o)f} = \frac{0.663}{(8/1)(0.02)(200{,}000)} = 20.7\ \mu\text{F}$$

The maximum ESR is determined from Eq. (6-36), using the maximum peak-to-peak variation in capacitor current. The peak-to-peak variation in capacitor current is the same as maximum inductor current. The average inductor current varies from 2.96 A at $V_s = 2.7$ V to 1.90 A at $V_s = 4.2$ V. The variation in inductor current is 0.762 A for $V_s = 4.2$ A, but it must be recalculated for $V_s = 2.7$ V using the 13.1-μH value selected, yielding

$$\Delta i_L = \frac{V_s D}{L f} = \frac{2.7(0.663)}{13.1(10)^{-6}(200{,}000)} = 0.683\ \text{A}$$

Maximum inductor current for each case is then computed as

$$I_{L,\max,2.7\text{V}} = I_L + \frac{\Delta i_L}{2} = 2.96 + \frac{0.683}{2} = 3.30\ \text{A}$$

$$I_{L,\max,4.2\text{V}} = I_L + \frac{\Delta i_L}{2} = 1.90 + \frac{0.762}{2} = 2.28\ \text{A}$$

This shows that the largest peak-to-peak current variation in the capacitor will be 3.30 A. The output voltage ripple due to the capacitor ESR must be no more than (0.02)(8) = 0.16 V. Using Eq. (6-36),

$$\Delta V_{o,\text{ESR}} = \Delta i_C r_C = I_{L,\max} r_C = 3.3 r_C = 0.16\ \text{V}$$

which gives

$$r_C = \frac{0.16\ \text{V}}{3.3\ \text{A}} = 48\ \text{m}\Omega$$

In practice, a capacitor that has an ESR of 48 mΩ or less could have a capacitance value much larger than the 20.7 μF calculated.

Inductor Resistance

Inductors should be designed to have small resistance to minimize power loss and maximize efficiency. The existence of a small inductor resistance does not substantially change the analysis of the buck converter as presented previously in this chapter. However, inductor resistance affects performance of the boost converter, especially at high duty ratios.